

# Hotel Industry Data Analysis

This report presents an in-depth analysis of the Indian hotel market, leveraging data from Hotel.ipynb.

Total Records Analyzed: 5,000 (Raw) 2,571 (Cleaned)

**Name:** Supriyo Maity

**Environment:** Jupyter Notebook



# Executive Summary & Objective

## Goal

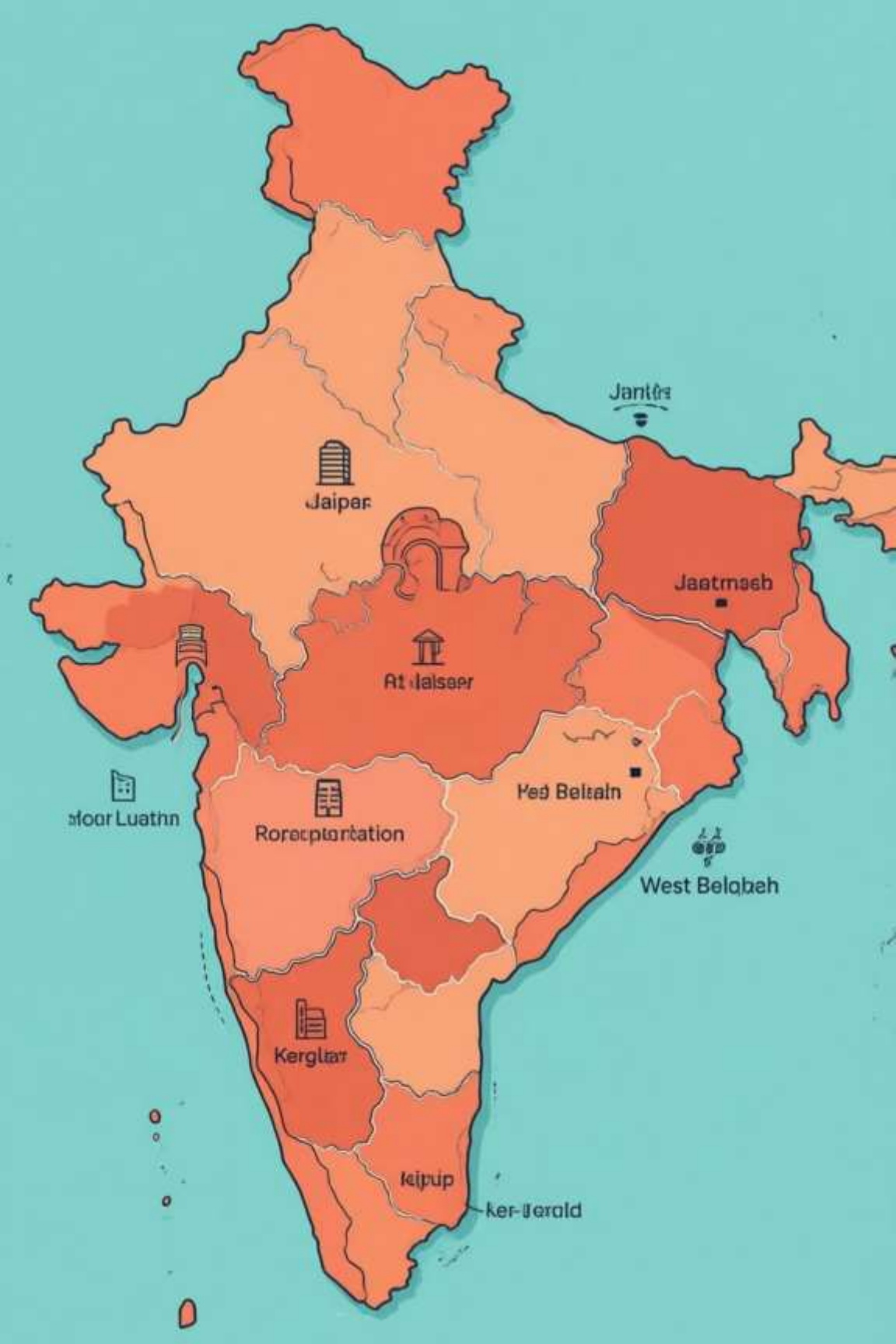
Analyze the Indian hotel market: inventory, quality ratings, and property types.

## Data & Methodology

Source: hotel\_analysis.xlsx. Python (Pandas, Matplotlib) for cleaning, aggregation, and visualization.

## Key Insight

Market dominated by budget accommodations (Avg Rating ~2.45). High inventory in Jaipur, Kerala, and West Bengal.



# Data Cleaning Pipeline

01

## Initial Data

5,000 rows with 15 columns.

02

## Column Reduction

Retained: Property Name, City, State,  
Star Rating, Room Count, Property Type.

03

## Star Rating Standardization

Parsed text (e.g., "3 Star hotel") to  
numeric (e.g., 3.0).

04

## Handling Nulls

Removed rows with missing data.

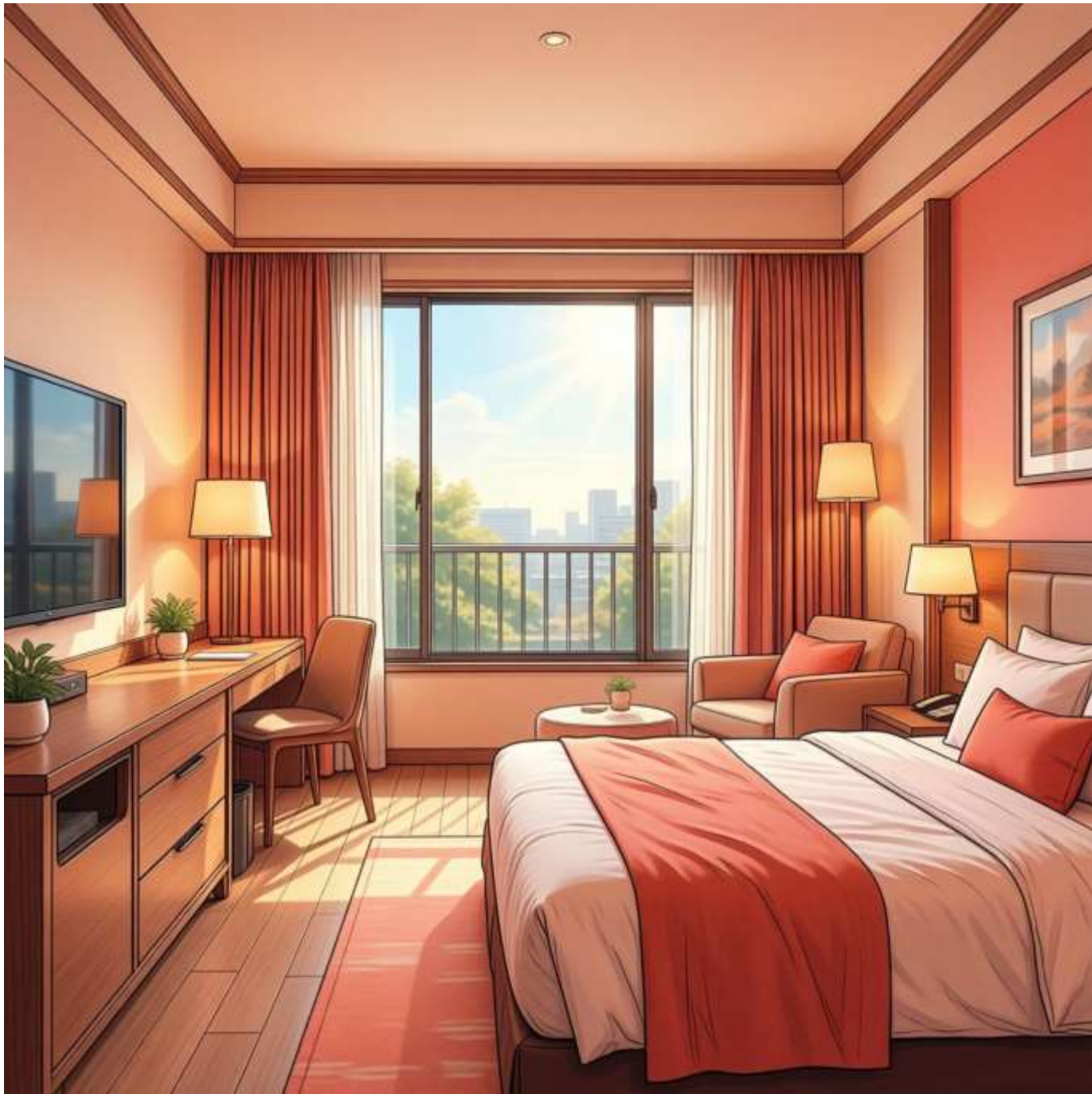
05

## Final Dataset

2,571 verified records for analysis.



# Market Overview: Descriptive Statistics



28.1

Avg. Room Count

Rooms per property

2.46

Avg. Star Rating

Out of 5 stars

14

25th Percentile

Most hotels are small

340

Largest Hotel

Max rooms

# The "Big Players": Market Leaders

Top 5 Properties by Room Capacity:



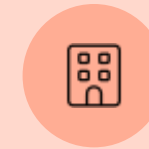
**Novotel Kolkata**

340 Rooms



**Jaipur Marriott**

303 Rooms



**Kochi Marriott**

274 Rooms



**Park Hyatt Goa**

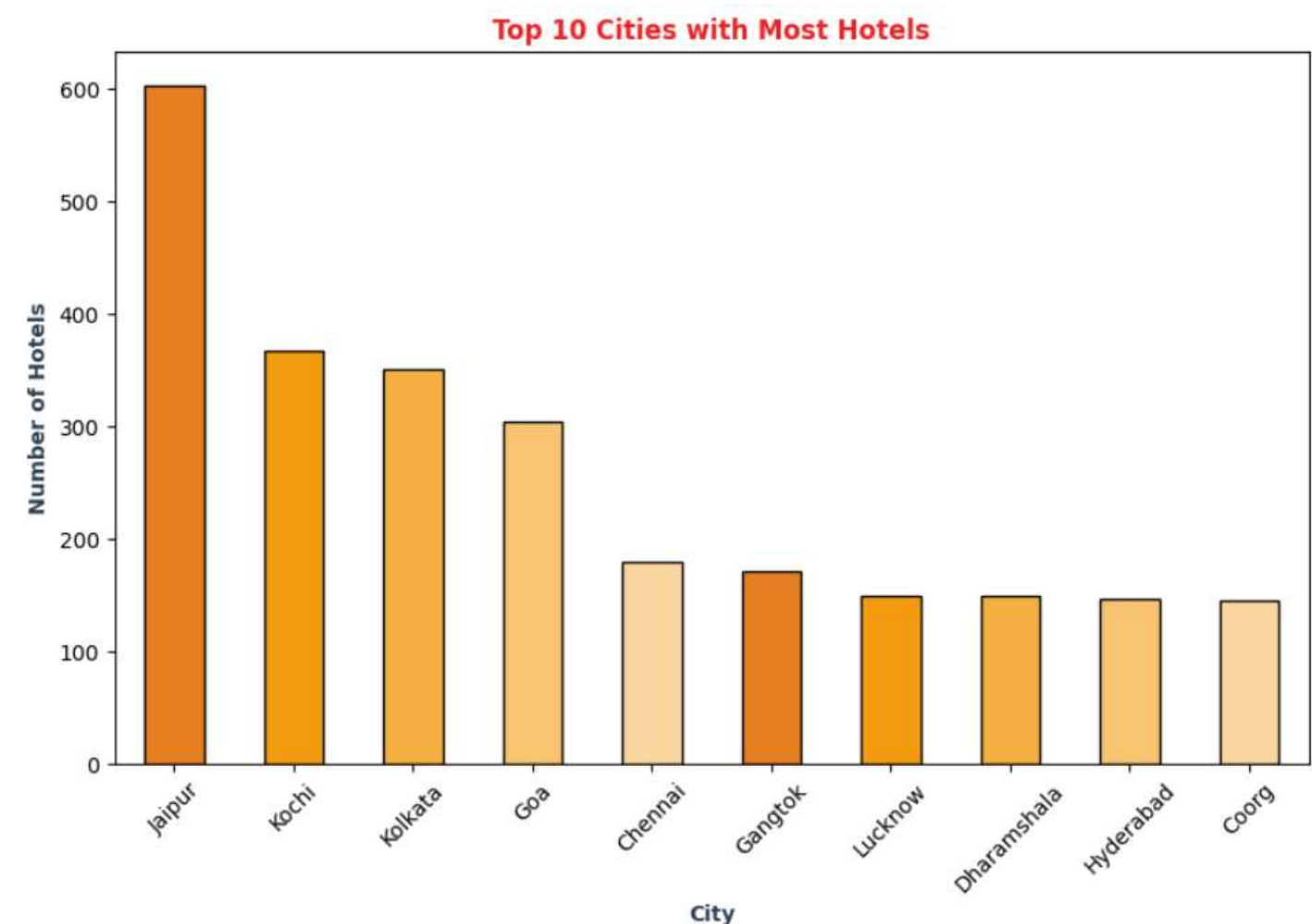
249 Rooms



**Estrela do Mar**

240 Rooms

# Geographic Supply: Top Cities

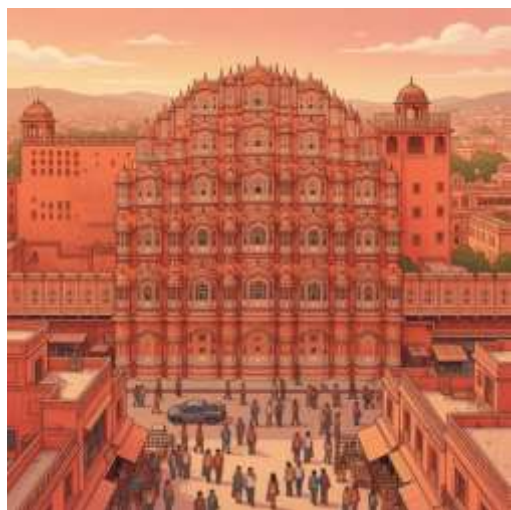


## Jaipur: Undisputed Leader

Jaipur leads in hotel density within this dataset.

## Observation

Tier-2 tourist hubs like Gangtok and Dharamshala are closely competing with metros like Kolkata in hotel volume.



# Regional Analysis: Top States

- 1

Rajasthan

436 Hotels (Driven by Jaipur's massive supply).
- 2

Kerala

249 Hotels (High density in Kochi).
- 3

West Bengal

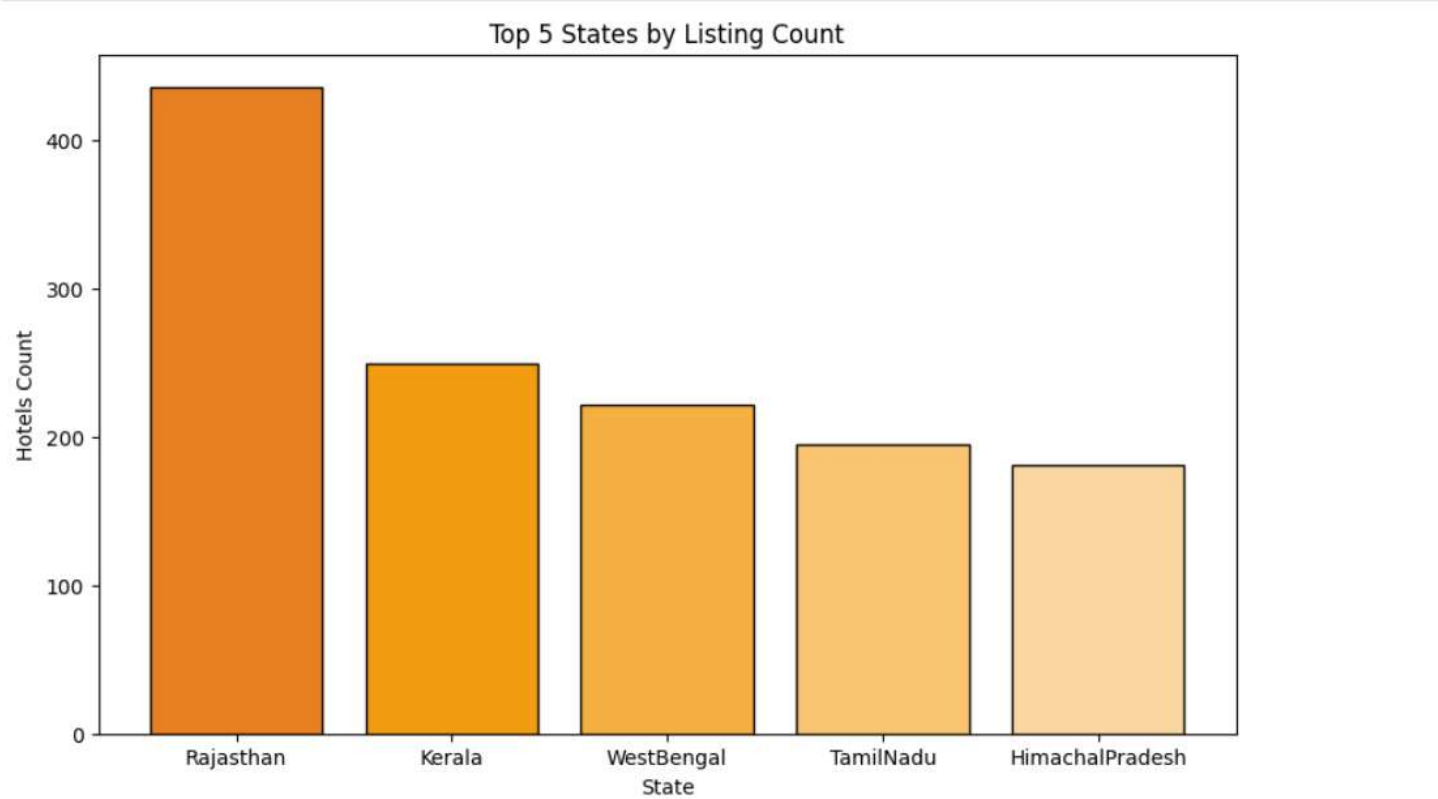
222 Hotels (Driven by Kolkata & Darjeeling region).
- 4

Tamil Nadu

195 Hotels.
- 5

Himachal Pradesh

181 Hotels.





# Quality by Property Type

An analysis of average user ratings reveals which accommodation types offer the best experience.

**Resort: 2.94**

Highest Rated Category

**Hotel: 2.41**

Average Rating

**Guest House: 1.60**

Lowest Rated





# Rating Disparities



**Resort**

Top-tier experience



**Hotel**

Solid, but not exceptional



**Lodge**

Similar to hotels



**Serviced Apartments**

Mid-range quality



**Homestay**

Significant drop in quality



**Guest House**

Lowest perception of quality



# The "Premium" Segment: True Luxury is Scarce

We filtered the data to find "True Luxury" using strict criteria, revealing a significant scarcity.

1

## Criteria Defined

Star Rating  $\geq 4.0$  AND Room  
Count  $> 50$ .

2

## Limited Count

Only 114 hotels qualified as  
"Premium."

3

## Small Market Share

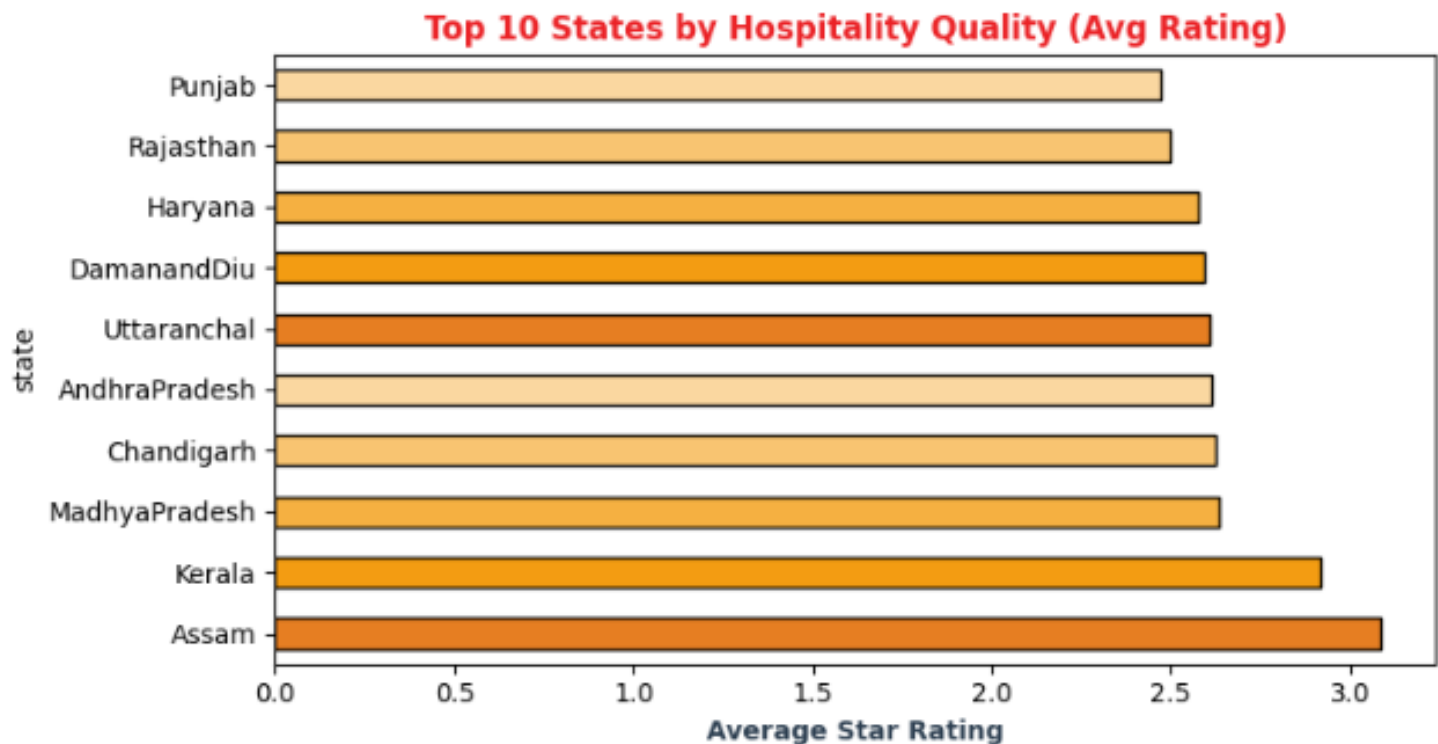
Represents just 4.4% of the total market.

# Geographic Quality: State-wise Analysis

We calculated the Average Star Rating per state.

High-quality, large-scale hospitality is a scarce commodity in this dataset. This highlights a significant gap in the market for premium offerings that meet stringent criteria.

"True luxury is not just about star ratings, but also scale and consistent service."



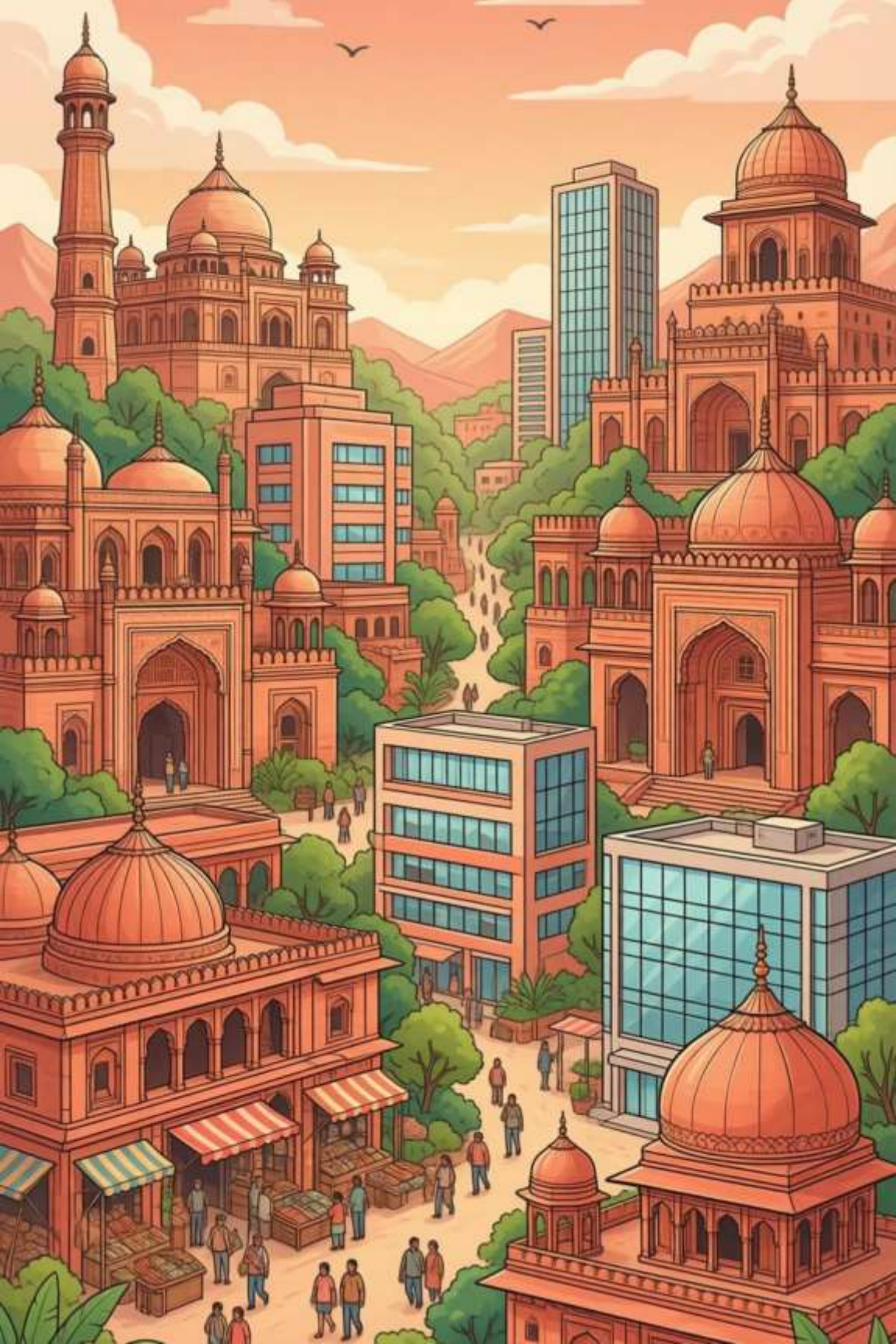
# Quality Over Quantity

This analysis identifies states that prioritize Quality over Quantity. While Rajasthan has the most hotels, states like Goa or Kerala often perform better on quality metrics than volume leaders.

📌 Focus on states with higher average ratings for potential investment in quality-driven hospitality.







# Conclusion & Recommendations

## Supply Saturation

Jaipur is highly saturated; new entrants should look at under-served markets.

## Quality Gap

Massive opportunity to upgrade "Guest Houses" and "Homestays" due to very low ratings (1.6 - 1.75).



# Investment Target: Resort Category

The "Resort" category commands the highest ratings (2.94), suggesting it is the most resilient business model for customer satisfaction.

Investing in high-quality resorts in under-served markets could yield significant returns and customer loyalty.



# Strategic Opportunities



## Explore New Markets

Identify and target regions with low saturation and high potential.



## Elevate Low-Rated Segments

Invest in improving quality of Guest Houses and Homestays to meet market demand.



## Focus on Resorts

Capitalize on the proven customer satisfaction and resilience of the resort model.



# Future Outlook

By addressing market gaps and focusing on quality, the hospitality sector can unlock new growth avenues and enhance overall customer experience.

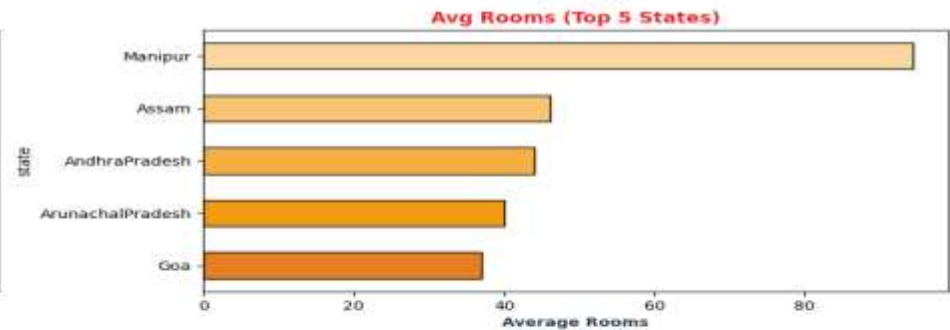
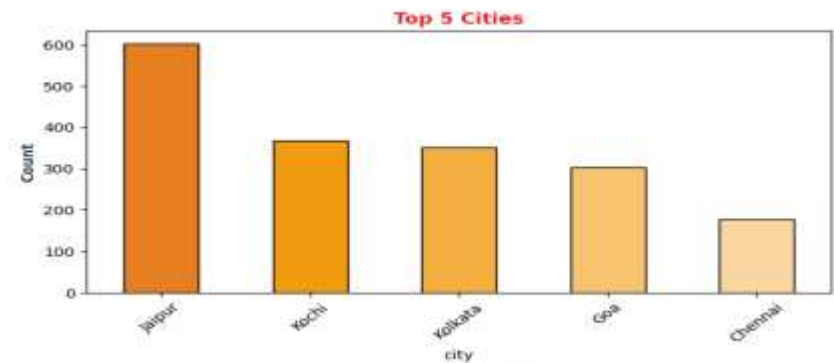
INNOVATION

QUALITY FOCUS

MARKET EXPANSION



Hotel Industry Analysis Dashboard



AC Status Based on Hotel Name

